

PROJECT DECLARATION

I hereby declare that this project report, titled “Customer Shopping Behaviour Analysis,” presents the analysis and dashboard work completed as part of my data analytics project. The report has been prepared using the project dataset, Python analysis, PostgreSQL data preparation, and Power BI dashboard.

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Date: _____

Signature: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during the completion of this data analytics project. I am thankful for the learning resources, guidance, and technical tools that helped me complete the data preparation, analysis, database work, and Power BI visualization. I also appreciate the opportunity to apply data analytics concepts to a practical customer shopping behaviour dataset.

Rajeswari K

ABSTRACT

This project presents an end-to-end analysis of customer shopping behaviour using Python, PostgreSQL, and Power BI. The analysis begins with data inspection and quality checking, followed by standardization and feature engineering. Age groups are created using quartiles, while purchase-frequency labels are converted into approximate day intervals. The prepared data is then loaded into PostgreSQL and visualized through an interactive Power BI dashboard. The final analysis highlights category revenue, customer demographics, subscription behaviour, discounts, seasonal performance, payment methods, and age-group purchasing patterns.

Keywords: Data Analytics, Python, Pandas, SQL, PostgreSQL, Power BI, Customer Behaviour, Data Visualization

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DATA ANALYTICS PROJECT REPORT

Customer Shopping Behaviour Analysis

Python • PostgreSQL • Power BI

Prepared by: Rajeswari K

Data source: Customer shopping behaviour dataset
Project Report

1. Executive Summary

This project analyzes customer shopping behaviour using a structured retail transaction dataset. The workflow covers data loading, data quality checking, preprocessing, feature engineering, relational data preparation with PostgreSQL, and interactive visualization in Power BI. The objective is to understand purchasing patterns, product-category performance, customer demographics, subscription behaviour, payment methods, seasonal trends, and age-group revenue.

The dataset contains 3,900 customer records. Total purchase value is \$233,081, with an average purchase amount of \$59.76 and an average review rating of 3.75/5. Clothing is the largest category by revenue, contributing \$104,264 (44.7%) of total purchase value.

2. Report Structure

- Introduction and project objectives
- Dataset and data dictionary
- Data cleaning and preprocessing
- Feature engineering
- Exploratory analysis and business insights
- PostgreSQL data preparation
- Power BI dashboard and visual analysis
- Recommendations, limitations, and conclusion

3. Introduction

Retail businesses generate large volumes of customer and transaction information. Analyzing this information can reveal which products generate the most revenue, which customer groups are most active, how customers prefer to pay, and whether subscription or discount behaviour is associated with purchasing activity. This project converts the raw customer shopping dataset into a cleaned analytical dataset and then presents the results through an interactive Power BI dashboard.

4. Problem Statement

The raw shopping dataset contains customer demographics, products, categories, transaction amounts, reviews, subscriptions, discounts, previous purchases, payment methods, and purchase frequency. Without structured preprocessing and visualization, it is difficult to identify the most important patterns and communicate them clearly. The project therefore aims to prepare the data and create an analytical view that supports customer and sales-related decision making.

5. Objectives

- Understand the structure and quality of the customer shopping dataset.
- Check for missing values and basic data consistency.
- Standardize column names for easier analysis.
- Create meaningful age-group and purchase-frequency features.
- Prepare the cleaned dataset for relational storage in PostgreSQL.
- Analyze revenue and customer distribution across categories and age groups.
- Study subscription, discount, payment, seasonal, and demographic patterns.
- Build a Power BI dashboard for interactive reporting and business communication.

6. Tools and Technologies

Table 1. Tools and Technologies

Tool	Purpose	Use in Project
Python	Data preparation and analysis	Pandas, CSV loading, profiling, feature engineering
PostgreSQL	Relational data storage	Prepared customer data stored in a customer table
Power BI	Interactive visualization	KPI cards, category charts, age-group analysis and filters
Jupyter Notebook	Development environment	Step-by-step data preparation workflow

7. Dataset Description

The dataset contains 3,900 rows and 16 original columns. It represents customer shopping behaviour with demographic, product, transaction, review, subscription, discount, payment, and purchase-frequency attributes.

Table 2. Dataset Data Dictionary

Field	Description
Customer ID	Unique customer identifier
Age	Customer age
Gender	Customer gender
Item Purchased	Purchased product
Category	Product category
Purchase Amount (USD)	Purchase value in US dollars
Location	Customer location
Size	Purchased size
Color	Product color
Season	Season of purchase
Review Rating	Customer review rating
Subscription Status	Whether the customer is subscribed
Discount Applied	Whether a discount was applied
Previous Purchases	Number of previous purchases
Payment Method	Payment method used
Frequency of Purchases	Reported purchase frequency

8. Data Understanding and Quality Check

Initial inspection was performed using the dataframe structure, descriptive statistics, and a null-value check. All 3,900 records were populated for the original 16 fields, and the notebook's missing-value check returned zero missing values for each field. The average customer age is 44.07 years, the average purchase amount is \$59.76, and the average review rating is approximately 3.75.

Table 3. Data Quality Summary

Metric	Result
Rows	3,900
Original columns	16
Unique customers	3,900
Average age	44.07 years
Average purchase amount	\$59.76
Average review rating	3.75/5
Missing values	0 across the original fields

9. Data Cleaning and Preprocessing

- Loaded the CSV dataset into a Pandas DataFrame.
- Inspected the first records and dataframe structure.
- Checked descriptive statistics for numerical and categorical fields.
- Checked missing values across all original columns; no missing values were present.
- Standardized column names to lowercase and replaced spaces with underscores.
- Prepared the dataset for downstream database storage and visualization.

10. Feature Engineering

Two analytical features were created to make the dataset easier to segment and visualize.

- Age Group: customer ages were divided into four quartile-based groups: Young Adult, Adult, Middle-aged, and Senior.
- Purchase Frequency Days: reported purchase frequencies were mapped to approximate day intervals. Weekly = 7, Fortnightly/Bi-Weekly = 14, Monthly = 30, Quarterly/Every 3 Months = 90, and Annually = 365.

The age-group feature supports demographic comparisons, while the frequency-days feature converts categorical frequency labels into an analyzable numerical representation.

11. Exploratory Analysis

The analytical results show a total purchase value of \$233,081. Category-level analysis indicates that Clothing is the leading category, followed by Accessories, Footwear, and Outerwear.

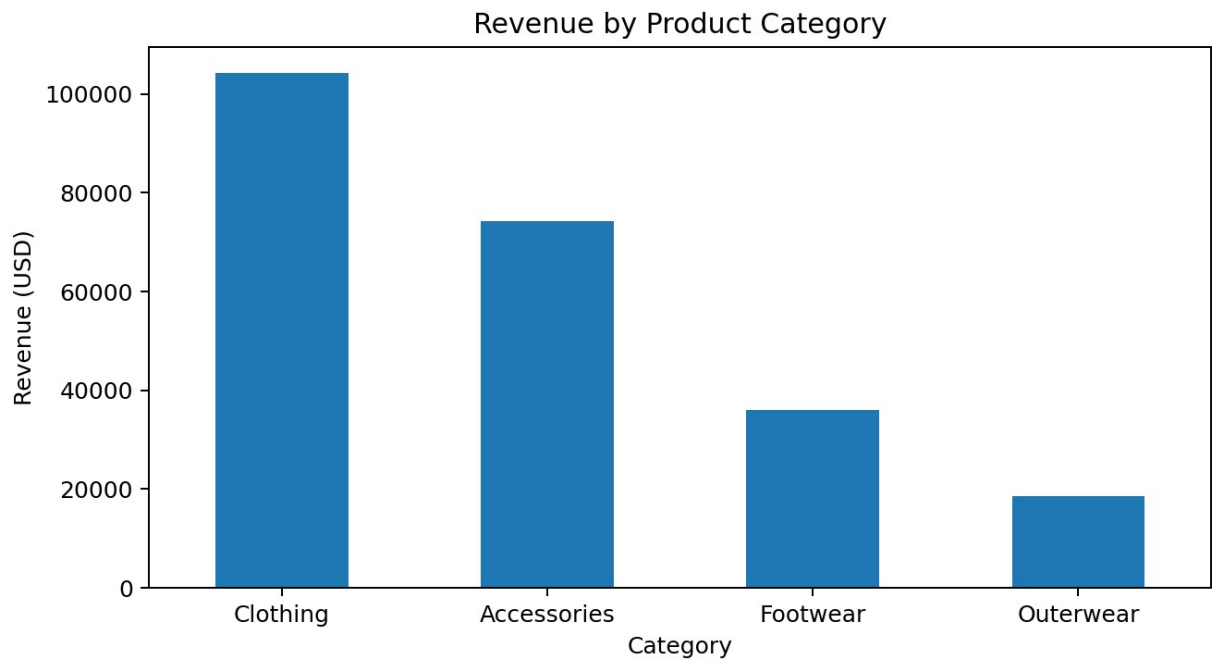


Figure 1. Revenue by product category

Clothing generated \$104,264, Accessories \$74,200, Footwear \$36,093, and Outerwear \$18,524.

12. Customer Demographic Analysis

The dataset is composed of 2,652 male customers (68.0%) and 1,248 female customers (32.0%). The mean customer age is 44.07 years, with ages ranging from 18 to 70.

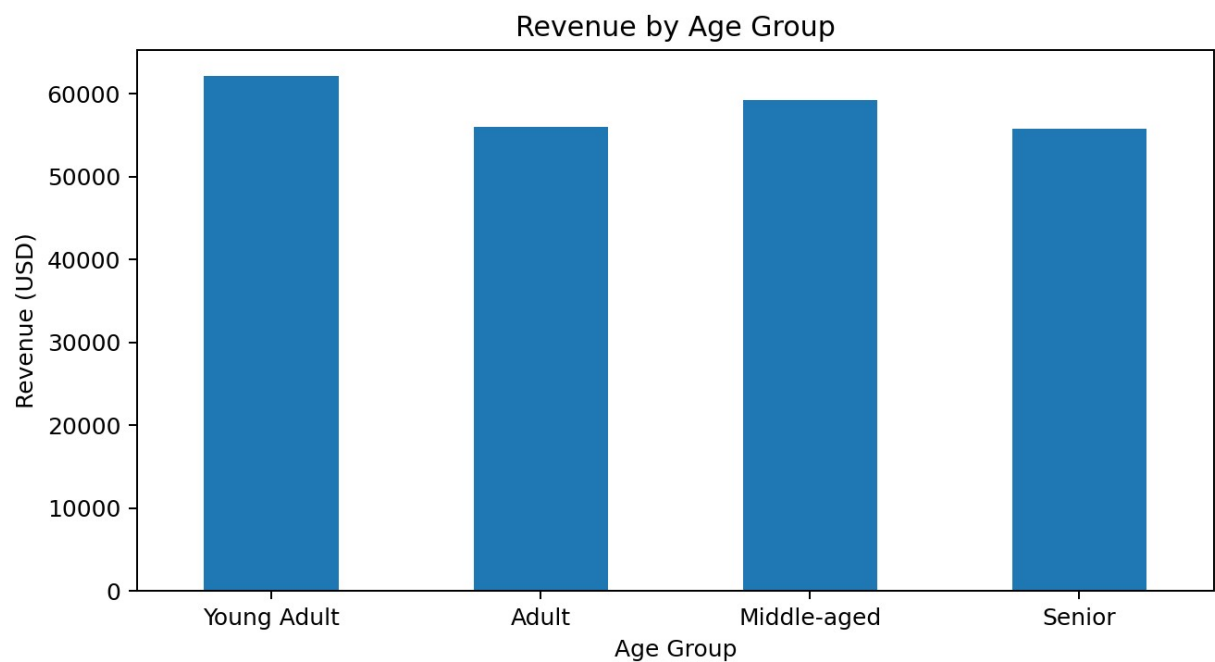


Figure 2. Revenue by age group

Young Adult customers generated \$62,143, the highest revenue among the four quartile-based groups. The average purchase amount is relatively close across groups, indicating that the main revenue differences are influenced more by the number of customers in each group than by large differences in average transaction value.

13. Subscription and Discount Analysis

There are 1,053 subscribed customers (27.0%) and 2,847 non-subscribed customers (73.0%). Discounts were applied to 1,677 purchases (43.0%).

In the dashboard, subscription status is exposed as an interactive filter and is also summarized through a percentage-of-customers visual. This makes it possible to compare the customer base by subscription status without changing the underlying dataset.

14. Seasonal and Payment Behaviour

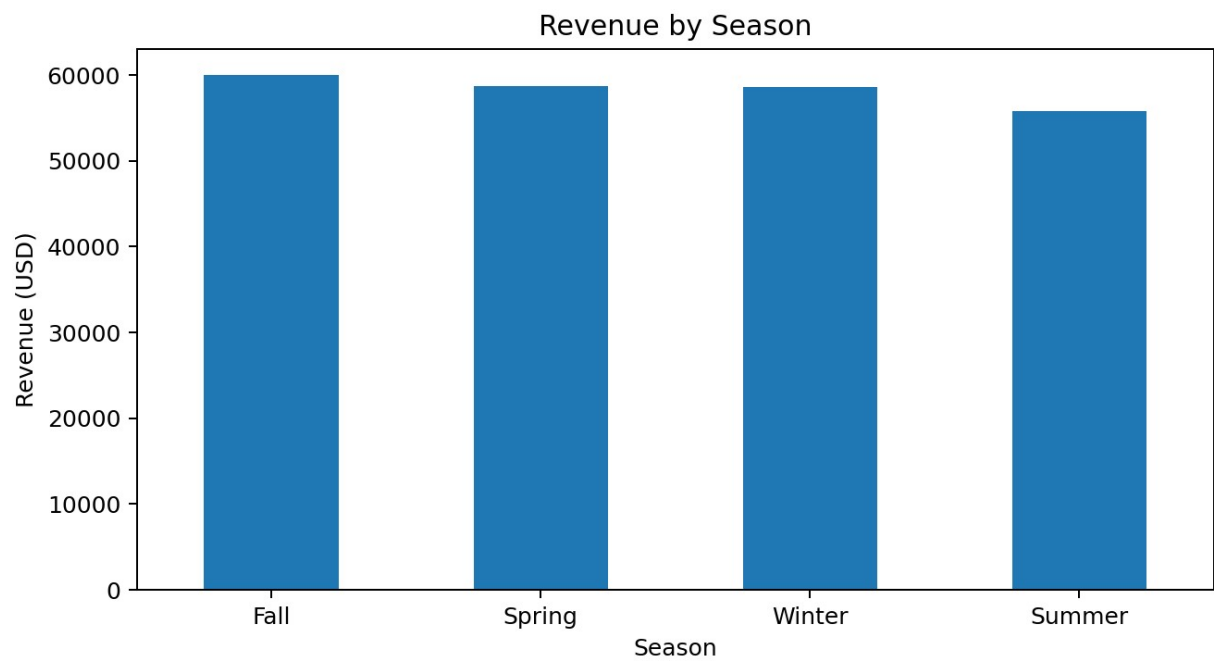


Figure 3. Revenue by season

Fall recorded the highest seasonal purchase value at \$60,018, followed by Spring (\$58,679), Winter (\$58,607), and Summer (\$55,777).

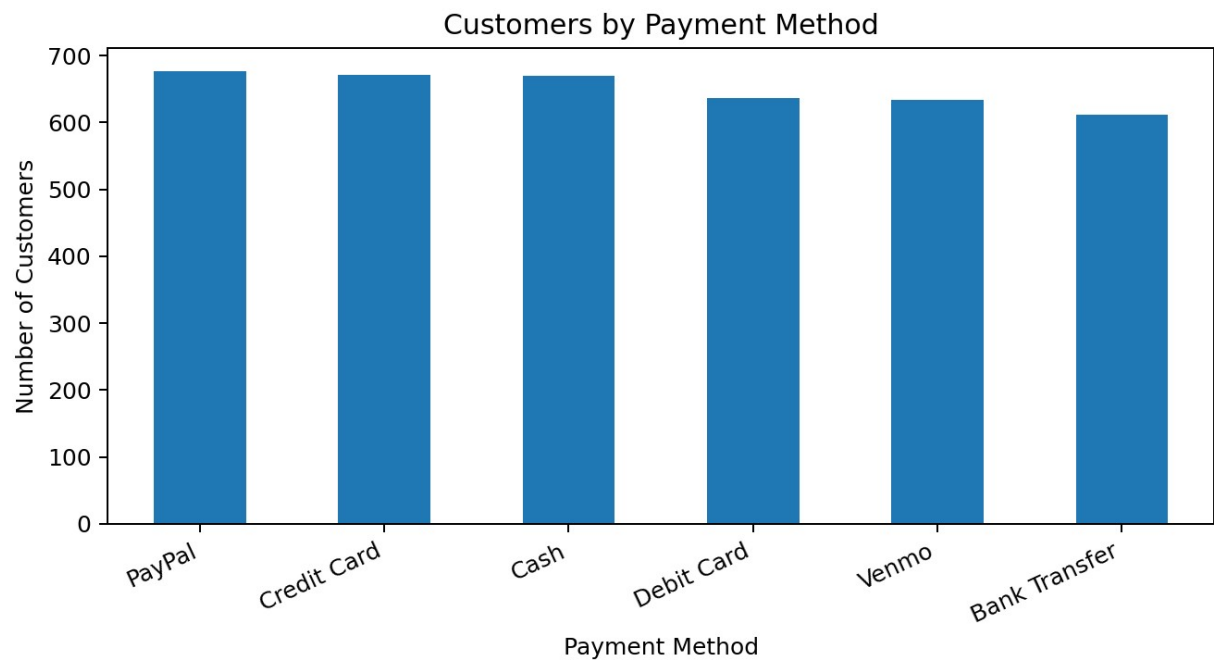


Figure 4. Customer count by payment method

Payment-method usage is relatively balanced across the six available methods. PayPal has the highest customer count at 677, followed closely by Credit Card and Cash.

15. PostgreSQL Data Preparation

After preprocessing, the dataframe was prepared for PostgreSQL storage. The project connects to a local PostgreSQL instance and writes the cleaned dataframe to a table named customer in the shopping_behavior database. This step demonstrates how the cleaned analytical data can be moved from a Python workflow into a relational database for structured querying and downstream reporting.

Note: the project materials provided for this report demonstrate the database-loading workflow. The report does not invent SQL query results that were not present in the supplied notebook.

16. Power BI Dashboard

The final Power BI dashboard, titled “Customer Behavior Dashboard,” provides an interactive summary of the customer shopping analysis. It combines KPI cards, slicers, a subscription donut chart, category-level revenue and sales visuals, and age-group comparisons.

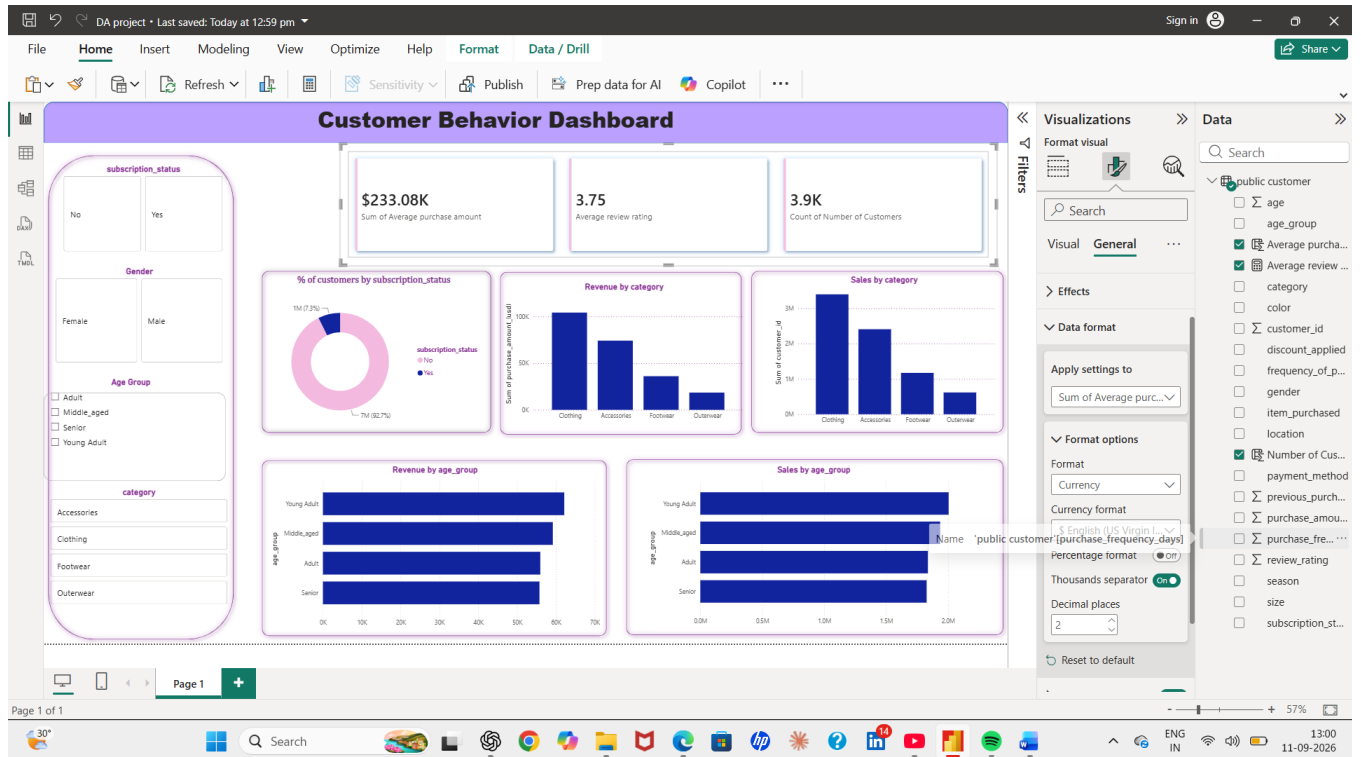


Figure 5. Final Power BI Customer Behavior Dashboard

17. Dashboard Components

Table 4. Power BI Dashboard Components

Dashboard Element	Purpose
KPI cards	Purchase amount, average review rating, and customer count.
Subscription status	Interactive status selection plus percentage distribution of customers.
Gender filter	Allows the dashboard to be filtered by gender.
Age Group filter	Allows comparison of Young Adult, Adult, Middle-aged, and Senior customers.
Category filter	Filters results by Clothing, Accessories, Footwear, and Outerwear.
Revenue by category	Compares purchase value across product categories.
Sales by category	Shows category-level customer/transaction volume as configured in the dashboard.
Revenue by age group	Compares purchase value across the four age groups.
Sales by age group	Shows the corresponding age-group sales distribution.

18. Key Findings

- The dataset contains 3,900 unique customer records and has no missing values in the original fields.
- Total purchase value is \$233,081, with an average purchase amount of \$59.76.
- Clothing is the strongest category, generating \$104,264, or 44.7% of total purchase value.
- Male customers represent 68.0% of the dataset, while female customers represent 32.0%.
- Only 27.0% of customers have an active subscription status of Yes in the dataset.
- Discounts were applied to 43.0% of purchases.
- Fall is the highest-revenue season in the dataset.
- Young Adult is the highest-revenue age group among the quartile-based groups.
- Payment method usage is distributed fairly evenly across the six payment options.

19. Business Recommendations

- Prioritize Clothing inventory and campaigns because it is the leading revenue category.
- Use age-group segmentation to design targeted offers, especially for the Young Adult segment.
- Investigate ways to increase subscription adoption because the subscribed group is substantially smaller than the non-subscribed group.
- Review discount effectiveness using purchase value and customer frequency before expanding discount campaigns.
- Prepare seasonal campaigns around higher-performing periods such as Fall while monitoring demand in lower-revenue seasons.
- Maintain multiple payment options because customers use a broad range of payment methods.
- Use the Power BI filters to monitor category, gender, age group, and subscription segments during business reviews.

20. Limitations

- The dataset represents recorded shopping behaviour and does not by itself establish causal relationships.
- The age groups are quartile-based, so their boundaries depend on the distribution of ages in this dataset.
- Purchase frequency was converted to approximate day intervals for analysis; this is a standardization choice rather than an observed exact number of days between transactions.
- The provided project materials do not contain a complete set of SQL analytical query outputs, so this report does not fabricate SQL findings.

21. Conclusion

The Customer Shopping Behaviour Analysis project demonstrates a complete practical data analytics workflow: loading and profiling a dataset, checking data quality, standardizing fields, creating analytical features, preparing the data for PostgreSQL, and presenting business-oriented insights through Power BI. The analysis identifies Clothing as the leading revenue category, highlights customer demographic differences, shows the relatively low subscription share, and identifies seasonal and payment-method patterns. The final dashboard converts these findings into an interactive format that can be used for business reporting and further exploratory analysis.

22. Future Scope

- Add SQL-based KPI and segmentation queries for deeper database-side analysis.
- Create customer-level RFM or loyalty segments if transaction history becomes available.
- Analyze the relationship between discounts, subscriptions, review ratings, and purchase value.
- Add time-series data when transaction dates are available to monitor trends over time.
- Publish the Power BI dashboard to a controlled workspace for stakeholder access and scheduled refresh.

23. References

1. Project dataset: shopping_behavior_updated (1).csv
2. Project notebooks: DA project.ipynb and supporting data analytics notebooks
3. Final Power BI dashboard: Customer Behavior Dashboard